TABLE XXIV. Minimum Requirements for the preparation of barge and tanker cargo tanks [1, 2]

LAST PRODUCT CARRIED (For information regarding 2nd and 3rd to last cargoes see note 8.)	PRODUCT TO BE LOADED							
	Gasolines[4,5]: MUR, MUM, MUP, EFM, E85, 100, 130	Jet Fuels: JAA/A1, JP8, F24, TS1, KS1/R/N	Jet Fuel: JP5	Jet Fuel: JPTS	Diesel Fuel: F76 Sulfur greater than 15 ppm and less than or equal to 15ppm	Diesel Fuels with B5 or less[5]: DF1, DF2, DS1, DS2, DSS, DSW, MGO	Biodiesel Blends greater than B5[5]: DF1, DF2, DS1, DS2, DSS, DSW	Heavy Fuel Oils: ASTM D396 (FS4, FS5, FS6) ASTM D975 (No. 4D) Residual Fuel Oils: ISO 8217, Class F (IFOs)
Gasolines[5]: MUR, MUM, MUP, EFM, E85, 100, 130, JAB, Naphtha, Reformate and Alkylate	A	A, B	A, B	F	A, B	A	A	A
Jet Fuels: JAA/A1, JP8, F24, TS1, KS1/R/N[9]	A	A	A, B	F	A, B	A	A	A
Jet Fuel: JP5	A	A	A	F	A, B	A	A	A
Jet Fuel: JPTS	A	A	A, B	A	A	A	A	A
Diesel Fuel (ULSD): F76 Sulfur less than or equal to 15 ppm	A, B	A, B	A, B	F	A	A	A	A, C
Diesel Fuel: F76 Sulfur greater than15 ppm	A, C	A, C	A, C	NO LOAD	A	A	A	A, C
Diesel Fuels with B5 or less[5]: DF1, DF2, DS1, DS2, DSS, DSW, MGO	A, C	A, C	A, C	NO LOAD	A, C	A, C	A	A, C
Biodiesel Blends greater than B5[5]: DF1, DF2, DS1, DS2, DSS, DSW	E	E	E	NO LOAD	A, C	A, C	A	A, C
Heavy Fuel Oils: ASTM D396 (FS4, FS5, FS6) ASTM D975 (No. 4D) Residual Fuel Oils: ISO 8217, class F (IFOs)	NO LOAD	NO LOAD	NO LOAD	NO LOAD	A,D	A,D	A,D	A
Crude[3]	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD	NO LOAD
Lube Oils	NO LOAD	NO LOAD	NO LOAD	NO LOAD	A, D	A, D	A, D	A
Water soluble chemicals[6]	A,C	A,C	A,C	F	A,C	A,C	A,C	A,C

Table XXIV continued, water soluble and/or water miscible chemicals

Product Name	Minimum Required Cleaning Procedures	Product Name	Minimum Required Cleaning Procedures
2-Butanone	A,C	Isopropanol, 91 %	A,C
1,4 butanediol	A,C	Isopropanol, Anhydrous	A,C
15W-50 (syn.hydrocarbons & additives)	A,C	Malaic Anhydride	A,C
2-Butanol	A,C	Maquat	A,C
2-Hydroxyethly Methacrylate (HEMA)	A,C	Methanol	A,C
Butendiol	A,C	Methyl Amyl ketone	A,C
CHRYSO	A,C	Methyl Pyrrolidone	A,C
Coralube 737 RB (= hexylene glycol)	A,C	Methyl Salicylate C8H8O3	A,C
Diethanolamine, IMO 9/ UN 3082/PG III	A,C	Monoethanolamine	A,C
Diethylene Trianine	A,C	Mpdiol Glycol	A,C
Diethylketone	A,C	MS (mineral sprint) rule 66	A,C
Dimethylacetamide (DML)	A,C	MTBE	A,C
Dipropylene Glycol	A,C	Eastman EB Solvent,	A,C
dipropylene glycol methyl ether (DPM)	A,C	Neodol 23-6.5	A,C
Diquat concentrate	A,C	N-Methyl-2-Pyrrolidone, Non Haz.	A,C
DOW 200	A,C	n-Propanol	A,C
DOW Froth 250	A,C	Petresul 550	A,C
Eastman Glacial Acetic Acid	A,C	Propyl Alcohol Sasol	A,C
Ethanol, Anhydrous	A,C	Propylene Carbonate, Non Haz.	A,C
Ethoxylated Sorbitan Laurate	A,C	Propylene Glycol Ethyl Ether (EPG)	A,C
Ethylene Carbonate, Non Haz.	A,C	Propylene glycol methyl ether (PM)	A,C
Ethylenediamine (EDA)	A,C	Terathane 2900 polyether glycol	A,C
Floquat (FL 4620 PWG)	A,C	Tetrahydrofuran (THF)	A,C
furfural alcohol	A,C	Triehtylene Glycol Ethyl Ether (ETG)	A,C
Furfurayl alcohol	A,C	triethanolamine 99	A,C
Gamma Butyrolactone	A,C	Triethylene Glycol	A,C
Glycerol	A,C	Tripropylene Glycol	A,C
Glycol Ether DB Acetate	A,C	UAN (Urea Ammonia Nitrate)	A,C
Glycol Ether PNP	A,C		
Hitec 614 (benzene sulfonic acid -calcium salt)	A,C		

TABLE XXIV continued, vessel cleaning requirements legend and notes

A	All cargo lines shall be dropped, tanks stripped, ballast residue removed.
B	All cargo and vent lines shall be drained of previous product and flushed with cold water. Cargo tanks must be thoroughly machine washed using cold water.Cargo tanks must be free of water, loose rust, sludge, mud, silt, etc.
C	The same as for "B," except that hot water shall be used instead of cold. If tank interiors are coated, water temperature should not exceed 58°C (136°F).
D	Cargo tanks and systems shall be processed in accordance with the instructions contained in EI HM 50 Guidelines for the cleaning of tanks and lines for marinetank vessels carrying petroleum and refined products.
E	Not to be loaded without special cleaning instructions. Three clean product/zero biological content intermediate cargoes required. (See Note 5 below)
F	Special tank preparations and cargo handling is required for JPTS, to prevent contamination. Tanks used for loading must be stainless steel or coated with an approved epoxy. Coating must be adherent: no flaking, peeling, or blistering. Prior to loading JPTS, tank cleaning requirements are: tanks must be machine washed with hot water, if cleaning chemical and/or salt water is used, the final wash must be with fresh water. Tank bottoms, interior bulk heads, and internalsmust be completely free of sediment, scale, and other contaminants. Tanks must be dry and all liquids completely removed from the tank's lines after cleaning,must be flushed with fresh water, drained and free of all water. Loading and unloading system must be completely isolated. This will be accomplished by completely separate piping systems or by use of blinds. Valves will not be depended on to effect isolation. Cargo tank valves will not be considered as part of the "double block" requirement for cargo system segregation. No common lines will be used. Steam smothering lines should have at least two valves that can be sealed from the main line to the tanks, or a blind installed that can be readily removed. Each tank will have its own individual vent. If ship has a common vent system, tanks used for JPTS must be isolated from balance of the vent system.

NOTES:

1. See Vessel Cleaning Requirements Legend for cleaning type code (A, B, C, D, E, F) details.
2. This table is included as a guide only. Requirements for tanker cleaning are determined by MSC vessel cleaning policy. Contact DLA Energy Quality Operations Division for tank cleaning requirements for any product not listed in the Table. If tank contains ballast water or water used for tank cleaning/washing, every effort shall be made to remove this free water to the minimum level possible by draining or by use of a pump.
3. There are no circumstances where a crude carrier is capable of cleaning tanks, pumps and lines sufficiently to load an aviation fuel immediately after a crude cargo.Crude carriers converted to a distillate diesel, aviation fuel or naphtha-based fuel must carry three cargoes of a commercial like product without a quality incident.
4. Gas free, lift scale and mop. Consideration may be given to substituting mopping by enhanced stripping and drying per EI (Energy Institute) HM 50 Guide.
5. FAME (Fatty Acid Methyl Ester) Contamination:

Aviation Gasoline or Aviation Kerosene/Turbine Fuel (Aviation Fuel Products)

- To avoid contamination from FAME in Aviation Gasoline, Aviation Kerosene/Turbine Fuel cargoes. See below instructions.

-- Following cargoes with pure biodiesel (B100) or any cargo with FAME content greater than 5 vol % (B5): Prior to loading Aviation Fuel Product cargoes requires three intermediate cargoes of a compatible fuel product. The three intermediate cargoes SHALL NOT contain FAME content greater than 5 vol % (B5). When any of the last three cargoes have consisted of diesel or heating oil products, FAME testing shall be conducted for each cargo; *attestation as described in the RFP is not sufficient for these products.* Examples of diesel or heating oil products include, but are not limited to, F-76, DF1, DF2, DS1, DS2, DSS, DSW, MGO, Gasoil, FS1, FS2, FS4, FS5, FS6, and IFO. Evidence by presentation of CoQs representing the last three cargoes with FAME testing (EN 14078 and ASTM D7371) demonstrating B5 or less shall be provided as proof. In scenarios where FAME testing (EN 14078 and ASTM D7371) was not conducted for a diesel or heating oil product, or such data is unable to be provided for all three of the last cargoes, the vessel shall be assumed to contain FAME greater than 5 vol %. The vessel tanks shall be evaluated and cleaned under the Minimum Requirements for Preparation of Tanker Cargo Tanks *table above* under procedures for Biodiesel Blends greater than B5 procedures. Alternatively, another vessel with FAME testing for the last three cargoes conducted may be nominated instead.

-- Tanks shall be in good condition and washing shall be particularly stringent, including flushing of pumps and lines, followed by drain and mop, at a minimum. A fresh- water rinse is required if sea water is used. Uncoated tanks shall be hot water washed and have loose bottom scale removed prior to loading an aviation fuel product cargo load.

-- Following cargoes with a FAME content of 5 vol % or less (B5 or below): Prior to loading Aviation Fuel Product cargoes, recommend a hot water wash, including flushing of the pumps and lines, followed by draining, at a minimum. A fresh-water rinse is required if sea water is used.

Special note:

Specifications for sensitive grades will require extreme care during sampling to avoid contamination from previous cargoes or buildup of residues. To reduce the possibility ofdrawing unrepresentative samples, cleaning of sampling equipment, vapor locks, standpipes and stilling wells is recommended as part of cleaning when tanks have previously held cargoes containing FAME.

6. Water soluble chemicals: include Ethanol, Mono Ethylene Glycol, Di Ethylene Glycol. See page four (4) for additional water soluble/miscible chemicals.
7. Previous automotive and aviation gasoline cargoes for loading JPTS are unacceptable unless the concentration of lead is verified to be less than 13ppm.
8. Certain products to be loaded on a tanker must not only consider the last cargo carried, but the 2nd and 3rd to last cargoes as well. These products include aviation turbine fuels and F-76 fuel oil, naval distillate. Previous cargoes of Crude oil and products containing FAME are addressed in notes 3 and 5. Other 2nd and 3rd to last cargoes of concern to include, but are not limited to, raw vegetable oil, lubricating oils, heavy fuel oils, and various chemicals. Determinations for acceptability of 2nd and 3rd to last cargoes carried prior to loading aviation fuel and F-76 will be made on a case-by-case basis using applicable Certificates of Analysis, Certificates of Quality and other analytical data.
9. Cargos carrying dyed kerosene shall use the A/B cleaning procedure.